Electronic Structure and Magnetic Anisotropy of MnTe Thin Films Revealed by Angle-Dependent XAS and XMCD

Jingjing Lu[1,2], Peng Zhang[1,2], Chen Li[1], Z.Q. Qiu[3], Vitaliy Lomakin[2], and Jeongmin Hong[1,3]

[1]School of Science, Hubei University of Technology, Wuhan 430068, China

[2]Center for Memory and Recording Research, University of California San Diego, La Jolla 92093, USA

[3]Department of Physics, UC Berkeley, CA 90230, USA

MnTe thin films with thicknesses of approximately 15 ~ 20 nm were grown on MgO substrates by molecular beam epitaxy and characterized using angle-dependent X-ray absorption spectroscopy (XAS) and X-ray magnetic circular dichroism (XMCD). The XAS spectra exhibit clear multiplet structures at the Mn L_3 and L_2 edges, indicating strongly localized Mn 3d states and confirming the high-spin $Mn^{2+}(3d^5)$ electronic configuration [1]. No significant changes in the XAS line shape or peak position are observed with varying measurement geometry, indicating a stable local electronic structure. In contrast, the XMCD signal shows a strong angular dependence, with the dichroic intensity gradually decreasing as the incident angle increases and becoming nearly negligible at large angles. The weak but finite XMCD response suggests the presence of uncompensated magnetic moments within the predominantly antiferromagnetic MnTe system. The observed angular dependence is attributed to variations in the projected magnetic moment along the X-ray propagation direction and reveals pronounced magnetic anisotropy. These results provide element-specific insight into the magnetic configuration of MnTe thin films and highlight their potential for antiferromagnetic spintronic applications.

***Index Terms*— XMCD; molecular beam epitaxy; magnetic compensation; altermagnet**

I. Introduction

THE eelectrical and magnetic control of spin states is central to next-generation spintronic memories, neuromorphic systems, and low-power logic devices. Magnetically compensated materials are particularly attractive because of their ultrafast spin dynamics, negligible stray fields, and stability against magnetic perturbations [2]. Unlike ferromagnets, compensated systems enable high-density integration with reduced crosstalk and Joule heating.

Among these materials, antiferromagnetic semiconductors combine semiconducting transport with rich spin physics. Manganese telluride (MnTe) is especially interesting due to its hexagonal structure, strong spin-orbit coupling, and non-collinear antiferromagnetic spin order [2,3]. Its canted spin configuration produces anisotropic magnetic responses and field-dependent magnetic compensation, while strong spin-lattice coupling enables potential control through strain, electric fields, and interfacial engineering.

Despite growing interest in MnTe, the relationship between its electronic structure and magnetic compensation remains unclear, particularly in epitaxial thin films relevant to device applications. In addition, the effect of magnetic field orientation on the compensated state has not been systematically explored using element-specific spectroscopy.

X-ray absorption spectroscopy (XAS) and X-ray magnetic circular dichroism (XMCD) are powerful tools for probing electronic and magnetic properties with elemental sensitivity [1]. In particular, angular-dependent XMCD can reveal magnetic anisotropy, non-collinear spin textures, and compensation behavior.

Here, we investigate Te-capped MnTe thin films grown by molecular beam epitaxy using Mn $L_{2,3}$-edge XAS and XMCD. The XAS spectra reveal a predominantly high-spin Mn^{2+}-dominated high-spin Mn^{2+} electronic state. The XMCD response shows strong angular dependence, including a distinct compensation regime where the dichroic signal nearly vanishes. The observed behavior is consistent with magnetic anisotropy and the presence of uncompensated magnetic moments in the antiferromagnetic MnTe system.

These results establish Te-capped MnTe thin films as a promising platform for studying field-direction-controlled magnetic compensation and tunable antiferromagnetic spin textures, with potential applications in low-power spintronic devices.

II. Methods

MnTe thin film samples with nominal thicknesses of ~15 to 20 nm were grown on MgO substrates by molecular beam epitaxy using elemental Mn and Te sources [4]. The growth was carried out at a substrate temperature of 300 °C under calibrated flux conditions, resulting in slight variations in thickness and stoichiometry among samples. To prevent oxidation, a Te capping layer with a thickness of ~2 nm was deposited in situ immediately after growth. XAS and XMCD measurements were performed at the Mn $L_{2,3}$ absorption edges using circularly polarized X-rays. Spectra were collected in total electron yield (TEY) mode under a constant magnetic field of 1.8 T. The external magnetic field was oriented at angles of 0°(out of plane), - 20°, - 40°, - 55°, and - 60° relative to the film normal, and data were recorded at room temperature using alternating-polarization scans.

III. Results and Discussion

Figure 1 presents the Mn $L_{2,3}$-edge XAS and XMCD spectra of the MnTe samples measured at different detection angles (0°~ -60°). All XAS spectra exhibit pronounced absorption features at approximately 635 ~ 637 eV and 646 ~ 649 eV, corresponding to the Mn L_3 and L_2 edges associated with the $2p_{3/2}\rightarrow 3d$ and $2p_{1/2}\rightarrow 3d$ electronic transitions, respectively. The spectra display clear multiplet structures, indicating the strongly localized nature of the Mn 3d electrons. Such spectral characteristics are consistent with the high-spin $Mn^{2+}(3d^5)$ electronic configuration commonly reported for MnTe systems [1]. Moreover, the

XAS line shape and peak positions remain nearly unchanged with varying measurement angles, suggesting that the local electronic structure and Mn valence state are stable under different measurement geometries. Therefore, the angular dependence mainly originates from changes in magnetic projection rather than modifications of the electronic structure.

In contrast, the XMCD spectra exhibit a pronounced angular dependence. A negative XMCD signal is observed near the Mn L_3 edge, while an opposite positive feature appears at the L_2 edge, which is characteristic of spin-polarized 3d transition-metal systems [1]. The overall XMCD intensity is relatively weak compared with typical ferromagnetic materials, indicating that the MnTe sample predominantly retains an antiferromagnetic nature with only a limited net magnetic moment. The finite XMCD response may arise from spin canting induced by the external magnetic field, interface exchange interactions, or defect-induced weak ferromagnetic components.

Since the XMCD signal reflects the projection of the magnetic moment along the X-ray propagation direction [1,5], the angle-dependent variation indicates that the magnetic moments preferentially align along a specific easy axis. A systematic reduction in XMCD intensity is observed with increasing measurement angle. The strongest XMCD signal appears at 0°, whereas the spectra measured at - 55° and - 60° become significantly weaker and nearly vanish. This behavior reveals the presence of strong magnetic anisotropy in the MnTe sample. Since the XMCD signal reflects the projection of the magnetic moment along the X-ray propagation direction, the angle-dependent variation indicates that the magnetic moments preferentially align along a specific easy axis. The fact that the XAS spectra remain essentially unchanged while the XMCD intensity varies strongly further confirms that the observed angular dependence originates primarily from magnetic orientation effects rather than chemical or electronic-state changes.

In addition, no obvious energy shift or spectral distortion is detected in the XMCD spectra at different angles, implying that the Mn valence state and local crystal-field environment remain unchanged throughout the measurements. Therefore, the angular variation mainly reflects modifications in magnetic configuration. Considering the known antiferromagnetic nature of MnTe, The observed finite XMCD signal may originate from uncompensated magnetic moments associated with spin canting, weak ferromagnetic contributions, interface-induced magnetic effects, or other symmetry-breaking mechanisms, potentially associated with lattice strain, structural defects, interface spin-orbit coupling, or Dzyaloshinskii - Moriya interactions.

Overall, the Mn $L_{2,3}$-edge XAS results confirm that Mn mainly exists in the high-spin Mn^{2+} state, while the XMCD measurements reveal a finite net magnetic moment and pronounced magnetic anisotropy. These results indicate that although the MnTe samples largely preserve the antiferromagnetic character, weak ferromagnetic responses and orientation-dependent magnetic behavior under external magnetic field and varying measurement geometries, providing important insight into the spin structure and magnetic anisotropy of the MnTe system.

IV. Conclusion

In summary, the angle-dependent XAS and XMCD measurements reveal that the MnTe sample maintains a stable Mn electronic configuration while exhibiting pronounced magnetic anisotropy. The Mn $L_{2,3}$-edge XAS spectra display characteristic multiplet features associated with the high-spin $Mn^{2+}(3d^5)$ state, and the absence of obvious spectral shifts with changing angle indicates that the local electronic structure and chemical valence remain unchanged. In contrast, the XMCD intensity strongly depends on the measurement geometry, with the signal gradually decreasing as the angle increases. This behavior demonstrates that the magnetic moments possess a preferred orientation, indicating the existence of a well-defined magnetic easy axis in the MnTe system. Although the overall XMCD signal is relatively weak, the finite dichroic response confirms the presence of a net magnetic moment, likely associated with uncompensated magnetic moments arising from spin canting, weak ferromagnetic contributions, or interface-related magnetic effects. These results suggest that the MnTe sample predominantly preserves its antiferromagnetic nature while exhibiting anisotropic magnetic behavior under external magnetic field conditions. The observed angular dependence provides important insight into the spin configuration and magnetic anisotropy of MnTe, which is highly relevant for understanding its potential applications in antiferromagnetic spintronic devices.

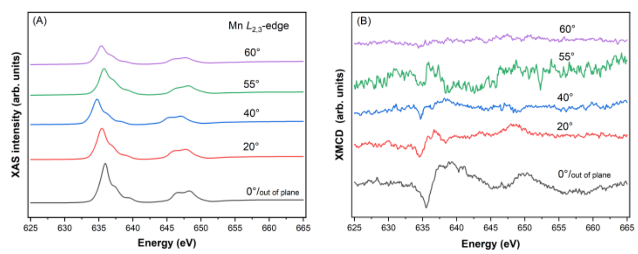

Fig 1. Angle-dependant (A) XAS and (B) XMCD spectra of Mn-Te samples

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